

Assignment No.: 1

1. Define **Machine Learning**. Explain its relationship with **Artificial Intelligence and Deep Learning** with suitable examples.
2. Explain the **types of Machine Learning**: Supervised, Unsupervised, Semi-Supervised and Reinforcement Learning with examples.
3. What is **Hypothesis Space**? Explain its concept with a suitable example.
4. What is **Inductive Bias**?
5. Explain the different **evaluation metrics for regression and classification** problems.
6. What is **Cross-Validation**?
7. Explain the **Decision Tree algorithm**. Describe its structure and working with a suitable example.
8. What is **Overfitting**? Explain its causes, effects and methods to prevent overfitting.
9. What is **Instance-Based Learning**.
10. What is **Feature Reduction**, Explain its need, advantages and commonly used techniques.
11. A student performance prediction system is given. Suggest suitable features, Machine Learning algorithm, evaluation method, and explain how overfitting can be avoided.
12. Explain **Collaborative Filtering-based Recommendation Systems** with a suitable real-world example.
13. Differentiate between **Linear Regression and Decision Tree** based on working, type of problem, advantages and limitations.
14. Explain the complete **Machine Learning model development and evaluation process**, starting from dataset preparation to final model testing.

15. Compare **Overfitting, Cross-Validation, Instance-Based Learning and Feature Reduction** and explain how each helps in building an effective Machine Learning model.